Adjoint-Faithful FNO Project Tracker

MITgcm baroclinic double-gyre benchmark

Status date: 23 July 2026 Canonical plan: `AF_FNO_Project_Plan.tex`

Current position. The control, low-wind, and high-wind trajectory datasets (S0–S2) are complete. S0 reproduces the expected qualitative tutorial circulation and has passed all seven numerical validation gates. A0, A, and B are frozen; B's 12-epoch realization completed with bitwise-identical reload and its 100-day evaluation shows materially better rollout than A but temperature and SSH still lose to persistence. Queued complete A/B characterization jobs 284816/284817 were cancelled with zero runtime. The immediate critical path is now forward-optimized Model C. Training-only C0 data/mode jobs 284850/284857 and C1a calibration job 285116 are complete. The clipped automatic weights were rejected, balanced coefficients were frozen from the unclipped gradient targets, pending GPU replica 284858 was cancelled, and training-only C1b job 285192 completed but was rejected because temperature/SSH increments remained 1.285/2.032 times persistence at the better learning rate. Training-only C1c duration job 285265 is running on the short CPU partition; no validation, inference, intermediate-wind, response, or adjoint data enter it. Model D response supervision and the complete adjoint campaign remain blocked until C passes the forward gate.

1 Status legend

Status	Meaning
Complete	Finished and checked against the stated acceptance evidence.
Active	Workstream is open and may contain completed substeps.
Next	Immediate executable step on the critical path.
Pending	Planned, but intentionally not started.

2 Milestone dashboard

ID	Milestone	Status	Evidence / exit condition
M0	Reproducible MITgcm build and segmented runner	**Complete**	S0 build job 283675; four-rank executable and immutable per-segment provenance.
M1	S0 control trajectory	**Complete**	100-year spin-up plus 10-year daily production; jobs 283676–283686; final data inventory verified.
M2	Tutorial validation	**Complete**	Validation job 283958; seven of seven gates passed; four project-facing figures and machine-readable report.
M3	S1/S2 forcing regimes	**Complete**	Low- and high-wind adjustment/production jobs 283959–283962 complete; daily inventories verified.

ID	Milestone	Status	Evidence / exit condition
R0	Git provenance baseline	**Complete**	Clean first commit pushed to the project remote; generated data, upstream clones, builds, environments, and scheduler logs excluded.
M4	Shared ML dataset	**Complete**	Jobs 284041/284042 built and independently validated the 5.3 GB S0–S2 Zarr store: $3 \times 3600 \times 46 \times 62 \times 62$, centered/masked state, split hashes, and training-only normalizers.
M5	A0 adapted Bire baseline	**Complete**	Jobs 284043/284050/284063/284065 completed the overfit, development, frozen model, and outputs. A0 improves held U/V forecasts but persistence remains better for temperature and SSH; metrics and figures are saved.
M6	A–D model ladder	**Active**	A0/A/B are frozen. Model B final job 284342 completed with exact reload, but mandatory temperature/SSH persistence criteria fail. Model C C0/C1a are complete; C1b rejected slow-field increment skill only, and one-variable C1c duration job 285265 is running. D is blocked until C passes the forward gate.
M7	I1/I2 wind interpolation	**Pending**	Two unseen intermediate-wind trajectories generated and opened for inference only.
M8	Forward-response campaign	**Pending**	224 ten-day perturbed restarts: 128 training, 32 validation, and 64 inference.
M9	Blind adjoint evaluation	**Pending**	Frozen artifacts/checksums recorded before 12 true-adjoint tests are generated or inspected.
M10	Control-update verification	**Pending**	Three forward checks demonstrate whether emulator-gradient controls improve each objective.

3 Completed simulation record

3.1 S0: control wind, $\tau_0 = 0.100\ \mathrm{N\,m^{-2}}$

Item	Recorded result
Execution	Build job 283675; spin-up jobs 283676–283685; production job 283686.
Model duration	100-year spin-up plus 10-year production: 110 model-years, using a 360-day year.
Production inventory	3,600 daily `dynState` records and 3,600 daily `surfState` records.
Measured size	4.6 GB in the raw MITgcm archive.
Validation	Job 283958; all seven tutorial gates passed.
Surface heat flux	Mean $-1.9589\ \mathrm{W\,m^{-2}}$.
Potential temperature	Final representative means: 17.6918°C at the surface, 8.3589°C at 305 m, and 2.4327°C in the abyss.
Gyre transport	First production-year extrema: +32.679 and −30.497 Sv.
Sea-surface height	First production-year range: −0.987 to +0.762 m.

The temperature structure, double-gyre transport, sea-surface-height pattern, and late-spin-up trends are consistent with the intended official tutorial behaviour. The deep ocean remains the slowest-adjusting component, as expected; this is recorded as a scientific caveat rather than a failed gate.

3.2 S1 and S2: off-control wind regimes

Run	τ_0	Jobs	Schedule	Daily output	Size
S1	$0.075\ \mathrm{N\,m^{-2}}$	283959 / 283960	5-y adjust + 10-y production	3,600 `dynState` + 3,600 `surfState`	2.7 GB
S2	$0.125\ \mathrm{N\,m^{-2}}$	283961 / 283962	5-y adjust + 10-y production	3,600 `dynState` + 3,600 `surfState`	2.7 GB

Both branches restart from the validated S0 year-100 state. Their five-year adjustment periods are kept separate from the ten-year daily production records used by the ML pipeline.

3.3 Retired high-resolution Bire reconstruction

The earlier 0.25° Bire-oriented spin-up, formerly Slurm job **283514**, was cancelled by the project owner on 21 July 2026. Its scratch tree, compiled build, MITgcm input deck, scheduler wrappers, conversion/training wrappers, and generated reports were subsequently deleted. This campaign is closed and is not a prerequisite for A0 or AF–FNO. Only recovered source and documentation that provide architectural evidence for A0 are retained.

4 Data locations and provenance

Asset	Canonical location
S0 project entry point	`outputs/af_fno/S0/`
S0 result catalog	`outputs/af_fno/S0/results_catalog.json`
S0 validation	`outputs/af_fno/S0/tutorial_validation/`
S0 provenance copies	`outputs/af_fno/S0/provenance/`
S0 raw archive	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/mitgcm/S0/`
S1 raw archive	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/mitgcm/S1/`
S2 raw archive	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/mitgcm/S2/`
Shared ML dataset	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v1.zarr`
Dataset manifest	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v1.manifest.json`
Dataset quality report	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v1.quality.json`
Pressure-postprocessor validation	`outputs/af_fno/pressure_validation_v1/pressure_validation.json`
A0 overfit artifacts	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A0/overfit_v1/`
A0 development artifacts	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A0/development_v1/`
A0 frozen artifacts	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A0/final_v1/`
A0 figures / numerical outputs	`outputs/af_fno/A0/final_v1/`
A0 paper-style output package (job 284150 complete)	`outputs/af_fno/A0/paper_style_v1/`
Model A overfit artifacts	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A/overfit_v1/`
Model A development artifacts (job 284152)	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A/development_v1/`
Model A frozen artifacts (job 284153)	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A/final_v1/`
Model A figures / numerical outputs (job 284155 complete)	`outputs/af_fno/A/final_v1/`
Model B overfit artifacts (job 284302 complete)	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/B/overfit_v1/`
Model B development profiles (jobs 284304–284306 complete)	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/B/development_*_v1/`
Model B frozen artifacts (job 284342 complete)	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/B/final_v1/`

Asset	Canonical location
Model B figures / numerical outputs	`outputs/af_fno/B/final_v1/`
Model C training-only diagnostics (jobs 284850/284857)	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/training_diagnostics_v1/` and `/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/training_diagnostics_v2/`
Model C GPU loss-gradient calibration (job 284858)	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/loss_calibration_v1/`
Model C CPU loss-gradient replica (jobs 284860/284864/285116)	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/loss_calibration_cpu_v1/`
Model C frozen loss manifest	`config/model_c_loss_v1.json`
Model C memorization/reload gate (job 285192)	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/overfit_v1/`
Model C controlled duration gate (job 285265)	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/overfit_duration_v2/`

The S0 project directory contains a symbolic link named `raw_mitgcm`; bulk output is not duplicated in `/home`. The catalog records counts, final pickup, hashes, executable provenance, and canonical paths. The shared-dataset manifest seals the three source archives, the fixed split, and normalizer hashes; its independent quality report records finite/mask/spectral checks, one-day increment ranges, and a fresh normalization recomputation.

5 Frozen project decisions

- The active scientific benchmark is the official 1°, $62 \times 62 \times 15$ baroclinic double gyre. Exact recovery of the paper's hidden 0.25° numbers is outside the primary claim; qualitative trends are the reproduction target.
- Model calendars use 360 days per year. The time step is 1,200 s, or 72 steps per model day.
- Standard trajectory production uses four MPI ranks. Long runs are split into restart-safe jobs that fit the two-hour `short` partition.
- S0–S2 share physics, grid, preprocessing, state definition, splits, and metrics. Only the prescribed forcing regime differs.
- True MITgcm adjoints are sealed test data. They are excluded from training, early stopping, architecture selection, hyperparameter tuning, perturbation-scale selection, and loss selection.
- All models must use the same immutable train/validation/test partition and training-only normalization statistics so architecture and loss comparisons remain interpretable.
- Git stores source, configuration, tests, small manifests, and project-facing documentation. Raw/reduced arrays, checkpoints, upstream clones, virtual environments, compiled trees, and scheduler logs remain external to Git.

6 Baseline and forward-selection ladder

Decision added 21 July and revised 23 July 2026. Establish an adapted Bire paper-architecture baseline, A0, before developing the modern NeuralOperator A–D sequence. This tests whether the published architectural idea recovers the paper's qualitative wind and forecast trends on the available tutorial configuration. Keep B frozen, select C only for forward fidelity, and add local-response supervision only in D.

A0 is a controlled adaptation, not a claim of exact historical reproduction. It keeps the recognizable Bire choices—channel-diagonal/separable spectral convolution, explicit pointwise channel mixing, three Fourier blocks, hidden width 128, positional encoding, and one-step state loss. Resolution-equivalent Fourier truncation uses 16 modes on the 62×62 grid. The frozen A0 did not use the paper code's stride-8 coarse pretraining; this is a recorded adaptation, not an inferred reproduction.

Changes forced by this benchmark must be disclosed: 46 Markov-state channels (centered U, V, temperature, and SSH), the 62×62 grid, ten production years per regime, wind stress as an input, and a ten-day primary comparison horizon. The obsolete public repository will be used as architectural evidence, not executed unchanged. Pressure is not a forgotten channel: evaluation contract v2 reconstructs MITgcm `PHIHYD` at levels 0, 7, and 14 from predicted temperature and SSH using the configured linear EOS and exact finite-difference hydrostatic operator. Its archived S0 validation passes with 3.89×10^{-4} $\mathrm{m^2\,s^{-2}}$ maximum error.

The comparison ladder is therefore:

$$\textbf{Persistence / climatology} \longrightarrow \textbf{A0: adapted Bire} \longrightarrow \textbf{A: modern state-only}$$
$$\longrightarrow \textbf{B: forward losses} \longrightarrow \textbf{C: forward-selected} \longrightarrow \textbf{D: AF–FNO response supervision}.$$

Every transition changes a declared element, making gains attributable rather than confounding the dataset, architecture, and derivative-aware objective at once.

7 Immediate execution sequence

Current next step: Model C stage C1c. The starting C architecture is B's dense four-layer, width-32 residual FNO, but its objective gives equal status to U, V, temperature, and SSH; adds training-only per-channel increment scaling; and compares tapered spectra of ten-day increments on the exact wet rectangle. The bounded later search contains modes $(12, 12), (16, 16), (16, 24), (24, 16), (24, 24)$, widths 32/64, and conditional padding/layer tests. Width 32 has 630,262 parameters; width 64 has 2,443,542 and is not justified unless capacity diagnostics require it. C0 jobs 284850/284857 found a 147-fold range in per-channel increment RMS, only 278–398 effective increment samples across regimes, and substantially more velocity-increment energy in meridional than zonal added modes. Complex-safe job 285116 completed in 4m39s. Its unclipped gradient targets froze increment/rollout/spectral/boundary coefficients at $0.001/0.15/10^{-5}/0.065$, producing weighted gradient ratios 0.423/0.495/0.207/0.245 relative to state. Pending redundant GPU job 284858 was cancelled. C1b job 285192 passed every criterion except increment skill: at the better learning rate, U/V were 0.254/0.442 times persistence but temperature/SSH were 1.285/2.032. Because its best checkpoint occurred at the final epoch, job 285265 changes only duration from 160 to 320 epochs at learning rate 0.0005. It remains a training-only gate, not a validated model.

1. Complete and interpret C1c job 285265. Require at least 90% total and 50% spectral reduction, every state-group error below 0.08, every increment group below persistence on the exact sampled records, rollout/boundary below 0.10/0.15, and bitwise-exact three-step reload.
2. Run bounded validation-only successive halving; do not open inference data until the selected architecture, schedule, three seeds, code, dataset, and normalizer hashes are frozen.
3. Run the complete C forward package. Generate I1/I2 and response data only after C passes the forward gate; train D only from the selected C design.

8 Remaining simulation and evaluation programme

Work package	Count	Role
I1/I2 trajectories	2	Intermediate wind amplitudes; inference-only interpolation test.
Perturbed forward responses	224	Ten-day local responses: 128 training, 32 validation, 64 inference.
Mandatory true adjoints	12	Three smooth objectives × two horizons (10/30 days) × two unseen states.
Forward control checks	3	Test one emulator-gradient control update for each objective.
Optional true adjoints	6	Ninety-day extension only if mandatory criteria are met.

The minimum programme contains 244 scientific integrations in total: five trajectory runs (S0, S1, S2, I1, I2), 224 perturbed forward responses, 12 mandatory adjoints, and three forward control checks. S0–S2 account for three completed trajectory runs.

The mandatory adjoints alone are the ground-truth gradient dataset. They evaluate: an interior Gaussian SSH average, a smooth western-boundary SSH average, and an area-weighted barotropic gyre-strength objective. The emulator, evaluation code, pass criteria, and cryptographic hashes must be frozen before these results are opened.

9 Risks, corrections, and controls

Risk / issue	Control or recorded correction
Scratch is not archival	Raw S0–S2 data live on `/bigscratch`. Copy them to durable project storage before the cluster retention deadline; retain catalogs, hashes, manifests, and figures under the project tree.
Trajectory-year arithmetic	The proposal's listed durations sum to **142**, not 132, model-years: $110 + 15 + 15 + 1 + 1 = 142$. Use 142 in compute/storage accounting.
Historical-code incompleteness	Label A0 "adapted paper architecture" and record all forced changes; do not claim a bitwise or numerical reproduction of Bire et al.
Data leakage	Freeze chronological indices and derive normalization only from training years. Keep I1/I2 inference-only.
Adjoint leakage	Enforce the sealed-adjoint contract and checksum frozen model and evaluation artifacts before generating or viewing adjoint results.
Deep-ocean drift	Report depth-dependent trend diagnostics with forecast results; avoid implying full abyssal equilibrium from upper-ocean stationarity.
Large job count	Pack short response integrations into auditable Slurm arrays or bundles while preserving one manifest/result record per scientific integration.
Cross-HPC portability	UCSB absolute paths, modules, accounts, and partitions are not NASA-portable. Introduce site profiles, rebuild dependencies and executables on the destination system, transfer data separately, and pass smoke/restart tests before production.

10 How to update this tracker

Update this document after a milestone changes state, a scientific decision changes, or a material dataset/model artifact is created. For each update:

1. advance the status date and change only milestones supported by durable evidence;
2. record Slurm job IDs, model-time duration, output counts, storage location, and validation result for every completed simulation;
3. add artifact paths and checksums where available; never treat a scratch pathname alone as an archival record;
4. record decision changes in both the relevant section and the change log;
5. regenerate `Project_tracking.pdf` and check the LaTeX log for undefined references, overfull boxes, or missing files.

Operational details and evolving status belong here. Scientific definitions, acceptance criteria, and the intended experimental design belong in `AF_FNO_Project_Plan.tex`; keep the two documents consistent without duplicating extensive derivations.

11 Change log

Date	Type	Change
2026-07-21	Tracker created	Consolidated S0–S2 execution evidence, validation metrics, data locations, active decisions, remaining simulations, and risks.
2026-07-21	Method decision	Added A0, an adapted Bire paper-architecture baseline, before models A–C.
2026-07-21	Accounting correction	Corrected total trajectory duration from 132 to 142 model-years.
2026-07-21	Infrastructure	Prepared the source tree for Git: removed disposable Slurm/LaTeX/Python artifacts, strengthened ignore rules, replaced the paper-only README with the AF–FNO project overview, and documented NASA portability. The first remote commit remains R0.
2026-07-21	Campaign retired	Recorded cancellation of job 283514 and removed the obsolete 0.25° MITgcm scratch data, build, input deck, job wrappers, and reports. Retained only Bire-derived code evidence required for A0.
2026-07-21	Dataset conversion	Added and locally tested the S0–S2 MDS-to-Zarr converter, including declared C-grid centering, wet-mask handling, chronological split codes, and training-only normalizers. All 39 available tests passed (one optional ML test skipped); Slurm job 284041 was submitted to create and validate `trajectories_v1.zarr`.
2026-07-21	M4 complete	Conversion job 284041 completed in 1m32s and produced the 5.3 GB $3 \times 3600 \times 46 \times 62 \times 62$ store. Independent job 284042 completed in 48s; it verified finite values, masks, split contract, spectrum, daily continuity, and recomputed the training-only normalizers to 3.12×10^{-7} mean and 1.56×10^{-7} scale maximum absolute error.
2026-07-21	A0 gate started	Added and smoke-checked the adapted A0 loader/trainer: 46 normalized state inputs, one normalized wind input, 46 normalized next-state targets, the recovered three-block separable FNO with 16 modes, masked paper-style MSE+MAE loss, and bitwise save/reload acceptance. Submitted the 96-sample GPU overfit gate as job 284043.
2026-07-21	A0 overfit passed	Job 284043 completed in 6m11s on one V100 GPU. Its 96 regime-balanced, training-only ten-day pairs reduced the masked normalized loss from 1.04999 to 0.002128 at learning rate 0.01; no fallback was needed, and saved/reloaded inference was bitwise identical. Checkpoint SHA-256: `30406885a1d4...295c83b0e9`.
2026-07-21	A0 development started	Added the sealed-split development runner with chunk-aware chronological batches and persistence comparison. Submitted V100 job 284050 for 12 epochs over 7,530 training pairs and 780 validation pairs; this is a development gate, not the frozen final A0 model.

Date	Type	Change
2026-07-21	A0 schedule frozen	Development job 284050 completed in 18m12s. Its best validation loss was 0.001841 versus persistence 0.014453, with bitwise checkpoint reload; the minimum occurred at epoch 10. Froze the final configuration (10 epochs, batch 8, learning rate 0.01, unchanged paper-style optimizer/loss/seed) and submitted its single final realization as V100 job 284063.
2026-07-21	A0 model frozen	Final job 284063 completed in 14m46s. It preserved bitwise checkpoint reload and obtained losses 0.001841/0.002272 on held validation/inference versus persistence 0.014453/0.014357. Checkpoint SHA-256: `b879b64b8cc7...3663be18f`. Submitted job 284065 to save the numerical and figure outputs in the project-facing output tree.
2026-07-21	A0 baseline closed	Evaluation job 284065 completed in 1m09s and saved `a0_metrics.json`, `a0_metrics_arrays.npz`, and three PNG figures under `outputs/af_fno/A0/final_v1/`. Held one-step A0/persistence RMSE ratios are 0.395 (U), 0.912 (V), 4.771 (temperature), and 13.612 (SSH); 10–100-day rollout figures show the same mixed state-variable result. M5 is complete without a claim that A0 beats persistence for every component.
2026-07-21	A0 paper-style evaluation complete	V100 job 284150 completed in 53s without retraining or changing A0 and saved a distinct output package. Its 15-member-per-regime, 10–200-day diagnostics adapt the forms of Bire et al. Figures 3, 5, 6, 7, and 9 to the 1-degree tutorial: multi-lead snapshots, spatial RMSE, ACC with training-global anomaly reference, and low/control/high wind-regime skill. This is explicitly a diagnostic-form comparison, not a numerical reproduction of the paper figures.
2026-07-21	Model A overfit submitted	After 48 passing project tests (one intentional dependency-path skip), style/diff checks, a 62-by-62 CPU forward/backward pass, and a production-Zarr sample read, submitted V100 job 284068. It ran the declared 96 regime-balanced, training-only pairs for the residual-learning/reload acceptance gate.
2026-07-21	Model A overfit passed	Job 284068 trained for 5m49s but failed only at checkpoint reload because NeuralOperator 2.0 adds a non-parameter `_metadata` entry that PyTorch rejects as a state-dict key. Filtered that entry from portable checkpoints, added a reload unit test, set `CUBLAS_WORKSPACE_CONFIG` for CUDA determinism, and reran the unchanged gate as job 284151. Job 284151 completed in 5m58s: the masked relative-L_2 loss fell from 0.52061 to 0.01589 (best), and save/reload was bitwise exact. Checkpoint SHA-256: `1a8ae303b797...43c21aeaad5b2`.
2026-07-21	Model A development submitted	After 52 passing project tests (one intentional dependency-path skip), submitted V100 job 284152. It trains only the 7,530 chronological train pairs and scores the 780 held validation pairs for 12 epochs, against persistence, to select a single final Model A training duration.

Date	Type	Change
2026-07-21	Model A schedule frozen	Development job 284152 completed in 17m24s. Its best held validation relative-L_2 loss was 0.018119 versus persistence 0.110743 at epoch 10; final epoch loss was 0.019212, and reload was bitwise exact. Froze one final schedule (10 epochs, batch 8, learning rate 0.001, AdamW $\beta = (0.9, 0.95)$, 10^{-5} weight decay, seed 20260721) and submitted V100 job 284153.
2026-07-21	Model A model frozen	Final job 284153 completed in 14m38s. It preserved bitwise checkpoint reload and achieved held validation/inference relative-L_2 losses 0.018119/0.018400 versus persistence 0.110743/0.110719. Submitted job 284155 to save Model A one-step, rollout, snapshot, and direct A0-versus-A figures plus numerical outputs in the project-facing output tree.
2026-07-21	Model A evaluation complete	Job 284155 completed in 1m38s and saved metrics/arrays plus four figures under `outputs/af_fno/A/final_v1/`. Held 10-day Model A/persistence RMSE ratios are 0.169 (U), 0.467 (V), 2.052 (temperature), and 2.277 (SSH), compared with A0's 0.395, 0.912, 4.771, and 13.612. At 100 days Model A still has temperature/SSH ratios 17.79/24.48; therefore A is a forward-state improvement over A0 but not a stable long-rollout solution.
2026-07-22	Model B gate submitted	Implemented the controlled Model B forward-loss pipeline without changing Model A's architecture, data, residual target, optimizer, or seed. The loss contract adds 20/30-day autoregressive error, 12-bin radial anomaly spectra at 10/20/30 days, and the first four wet cells from the western wall at all three leads. Added incremental development profiles, frozen-protocol A0/A/B/persistence evaluation, loss-contract hashing, checkpoint reload gates, tests, and Slurm wrappers. All 62 available tests passed (one intentional skip), and a production-grid three-step CPU forward/backward check had finite gradients. Submitted the 96-sample full-loss V100 acceptance run as job 284302.
2026-07-22	Model B overfit passed	Job 284302 completed in 13m12s. On 96 balanced training-only rollouts it reduced complete loss from 1.52456 to 0.04213 and one-step state loss from 0.52075 to 0.01704; rollout, spectral, and western-boundary components all fell, learning rate 0.001 passed without fallback, and checkpoint reload was bitwise exact. Checkpoint SHA-256: `512707d82bce...17871c159e28`. Dependency-gated rollout-only, rollout-plus-spectral, and full-loss development jobs 284304–284306 started after acceptance.

Date	Type	Change
2026-07-22	Model B schedule frozen	Development jobs 284304–284306 completed in 43–44 minutes with bitwise reload. At each profile's held minimum, common full-objective values were 0.044899 (rollout only), 0.044081 (rollout plus spectra), and 0.043064 (full); the full profile also had the lowest state (0.017925), rollout (0.033599), spectral (0.009012), and boundary (0.031558) terms. Its minimum was epoch 12 versus persistence 0.306148. Froze epochs 12, batch 8, learning rate 0.001, unchanged optimizer/seed/loss weights, and submitted final job 284342. Full-development checkpoint SHA-256: `e185477b126c...af23a7a8119d`.
2026-07-23	Model B closed	Final job 284342 completed in 36m19s on one V100, preserved bitwise reload, and saved the frozen checkpoint. Evaluation job 284346 completed the common 10–100-day package. B reduces A's 100-day U/V/temperature/SSH errors to 0.521/0.409/0.450/0.405 of A, but its own persistence ratios remain 0.449/0.997/8.009/9.925. B is retained unchanged as a successful rollout-loss ablation, not accepted as the forward parent of the response model.
2026-07-23	Complete jobs cancelled	Complete one-year A/B characterization jobs 284816 and 284817 were pending with zero runtime and were cancelled. Their frozen checkpoints and existing evidence remain; uniform final packages are deferred until C freezes so they are generated once for the final comparison.
2026-07-23	Model ladder revised	Inserted forward-selected Model C between B and the response-supervised model, now named D. The Duruisseaux practical FNO workflow and Bire evidence pattern remain the references. C uses a bounded dense-FNO search and must pass the complete forward gate before response or adjoint work opens.
2026-07-23	Model C C0 started	Added the bounded C architecture/loss contract and training-only diagnostics. Equal U/V/temperature/SSH state and boundary terms, per-channel ten-day increment RMS, and Hann-tapered increment spectra directly address B's aggregate-loss and spectral weaknesses. Eight Model C tests, style checks, shell validation, and a 62×62 three-step forward/backward pass succeeded; width-32/64 parameter counts are 630,262/2,443,542. CPU job 284849 exposed an incorrect assertion that checked future pair-start codes rather than target snapshot codes and stopped before creating output. Added a regression test and submitted corrected job 284850 with unchanged train-only scope.

Date	Type	Change
2026-07-23	Model C C0 complete	Corrected job 284850 completed in 42s and verified the immutable dataset hash and train-only read contract. Its 96-sample audit found a 147-fold increment-RMS range and only 278–398 effective increment samples. Because extra meridional modes retained more velocity-increment energy than extra zonal modes, added $(24, 16)$ before validation and reran job 284857 in 33s. At $(16, 16)$, U/V/temperature/SSH increment fractions are 8.8/24.0/67.8/85.6%; at $(24, 16)$, 12.5/32.9/70.2/89.0%.
2026-07-23	Model C C1a submitted	Added a training-only loss-gradient calibration: 20-epoch core warmup on 96 balanced rollouts, followed by raw-term and parameter-gradient audits. It proposes, but does not freeze, auxiliary weights and writes no candidate checkpoint. Nine Model C tests, style, shell, and production-grid checks passed; submitted V100 job 284858.
2026-07-23	CPU C1a replica corrected	CUDA is not scientifically required for the 96-rollout calibration. Job 284860 failed safely after 19s, before output, because its initial audit squared very large gradients in float32. Moving the global norm to float64 prevented overflow, but live job 284864 then revealed that a direct cast discarded imaginary spectral-weight gradients; it was cancelled before output. The final helper preserves real gradients in float64 and complex gradients in complex128 and is covered by a large real/complex regression test.
2026-07-23	Model C C1a complete	Complex-safe CPU job 285116 completed in 4m39s using 1.53 GB maximum RSS. After warmup, state/increment/rollout/spectral/boundary gradient norms were 1.965/830.5/6.484/40,763/7.411. Rejected the clipped 0.01 increment/spectral proposal and froze rounded unclipped coefficients $0.001/0.15/10^{-5}/0.065$, whose weighted gradient ratios to state are 0.423/0.495/0.207/0.245. Report SHA-256: `3406c9104dd6...592e469`. Cancelled redundant pending GPU job 284858 with zero runtime.
2026-07-23	Model C C1b submitted	Added a 96-rollout, training-only memorization gate with fixed initialization/batch order across the bounded learning-rate fallback. Acceptance is groupwise rather than aggregate: 90% total reduction, 50% spectral reduction, maximum state group 0.08, maximum increment group 1.0, rollout 0.10, boundary 0.15, and bitwise-exact three-step reload. Eleven Model C tests and all style/diff/shell checks passed; submitted eight-core short-queue job 285192.

Date	Type	Change
2026-07-23	Model C C1b rejected	Job 285192 completed both 160-epoch attempts in 58m30s and failed only the increment criterion. At learning rate 0.0005, total/spectral reductions were 97.7/99.1%, maximum state group was 0.0304, rollout/boundary were 0.0312/0.0397, and U/V increments were 0.254/0.442 times exact-subset persistence; temperature/SSH remained 1.285/2.032. The absolute-1 threshold is replaced by the mathematically correct per-group ratio to persistence on the same sampled records, without relaxing the requirement.
2026-07-23	Model C C1c submitted	The lower learning rate reached its best total at epoch 160, so undertraining is tested before changing objective or capacity. Added version-2 diagnostics that preserve full learning curves and a reload-tested best checkpoint on rejection. Job 285265 fixes data, architecture, loss, seed, batch order, and learning rate 0.0005, changing only duration from 160 to 320 epochs.
2026-07-23	Pressure evaluation completed	Added a fixed MITgcm-consistent `PHIHYD` postprocessor without changing any model input, output, weight, or split. It integrates the configured linear-EOS density anomaly and adds the linear-free-surface $g\eta$ term. Against the archived S0 `PH` dump at iteration 2,851,200, all-depth wet-cell maximum error is 3.89×10^{-4} and RMSE is 9.75×10^{-6} $m^2\,s^{-2}$. Evaluation contract v2 adds surface/mid/bottom pressure curves, maps, snapshots, spectra, boundary metrics, and gate comparisons. Existing v1 packages remain immutable and pre-pressure; frozen inference will be replayed once for the uniform v2 comparison after C freezes.